

First Normal Form (1NF) -- Relational Database Design

Definition: A relation is in First Normal Form (1NF) if and only if all attributes contain only atomic (indivisible) values, and there are no repeating groups or arrays stored within a single field.

1. Detailed Technical Explanation

Normalization is the systematic process of organizing relational database schemas to eliminate data redundancy and anomalies (Insertion, Deletion, and Update anomalies).

Key Rules of 1NF:

1. Atomic Values: Each column must store a single value (no multi-valued attributes like comma-separated phone numbers).

Conversion Example: Unnormalized Table to 1NF

Unnormalized Relation (UNF):

101	Rahul Sharma	9876543210, 9123456789	DBMS, DSA, MPCA
102	Anita Verma	9988776655	DBMS, Python

Normalized 1NF Relation:

101	Rahul Sharma	9876543210	DBMS
101	Rahul Sharma	9876543210	DSA
101	Rahul Sharma	9876543210	MPCA
101	Rahul Sharma	9123456789	DBMS
101	Rahul Sharma	9123456789	DSA
101	Rahul Sharma	9123456789	MPCA
102	Anita Verma	9988776655	DBMS
102	Anita Verma	9988776655	Python

2. Core Concepts & Memory Keywords

- * Atomic Domains: Fields storing single indivisible values.

3. Must-Write Points for Exams

- * 1NF eliminates multi-valued attributes and composite attributes from relation instances.

4. Quick Recall Flow

Unnormalized Table -> Remove Comma-Separated Values -> Ensure Atomic Cells -> 1NF Form Achieved